

Lab # 13

Structure in C++

Objective

The students will learn and apply the

Introduction

In C++, a **structure (struct)** is a user-defined data type that allows grouping variables of different data types under a single name. Structures are useful when dealing with complex data where multiple attributes are required to represent an entity, such as a student record or employee details.

Theory

A structure is defined using the struct keyword. It can contain variables of different types such as integers, floats, and characters.

Syntax

```
struct StructureName {  
    dataType member1;  
    dataType member2;  
};
```

- Each variable inside the structure is called a **member**.
- Structure variables can be accessed using the **dot (.) operator**.

Conclusion

In this lab, we learned how to use **structures in C++** to store and manage multiple related data items. Structures improve program organization and make handling complex data easier.

Lab 13 Tasks

1. Structure Declaration and Initialization

- Declare a structure called Person with the following members:
 - name (string)– age (integer)
 - address (string).
- Declare a variable called person1 of type Person and initialize its members with values of your choice.
- Display the values of the members of person1 using dot notation.

Code:

```
#include <iostream>
using namespace std;

// Define a structure named Person
// Yeh structure Person ke naam se banaya gaya hai
struct Person {
    string name;    // Person ka naam store karega
    int age;        // Person ki age store karega
    string address; // Person ka address store karega
};

int main() {
    // Declare and initialize structure variable
    // Yahan hum person1 naam ka variable bana rahe hain aur values assign kar rahe hain
    Person person1 = {"Ali", 20, "Rawalpindi"};

    // Display the values using dot notation
    // Dot operator use karke values print kar rahe hain
    cout << "Name: " << person1.name << endl;    // Name print karega
    cout << "Age: " << person1.age << endl;        // Age print karega
    cout << "Address: " << person1.address << endl; // Address print karega

    return 0; // Program end
}
```

2. Structure Declaration and Initialization

- Declare a structure called Student with the following members:

– name
(string)– age
(integer)
– grade (char)

• Declare a variable of type Student called student1 and initialize its members with values of your choice.

- Display values of student1 structure members.

Code:

```
#include <iostream>
using namespace std;

// Define a structure named Student
// Yeh structure Student ke naam se banaya gaya hai
struct Student {
    string name; // Student ka naam store karega
    int age;     // Student ki age store karega
    char grade;  // Student ka grade store karega
};

int main() {
    // Declare and initialize structure variable
    // Yahan hum student1 naam ka variable bana rahe hain aur values assign kar rahe hain
    Student student1 = {"Ahmed", 19, 'A'};

    // Display the values using dot notation
    // Dot operator use karke values print kar rahe hain
    cout << "Name: " << student1.name << endl; // Name print karega
    cout << "Age: " << student1.age << endl;    // Age print karega
    cout << "Grade: " << student1.grade << endl; // Grade print karega

    return 0; // Program end
}
```

3. Accessing Structure Elements

- Declare a structure called Book with the following members:
 - title (string)
 - author (string)– price (float)
 - pages (integer)
- Declare a variables of type Book called book1 and book2, and initialize its members with values of your choice.
- Display values of structure members of variables book1 and book2 .

Code:

```
#include <iostream>
using namespace std;

// Define a structure named Book
struct Book {
    string title;    // Book ka title store karega
    string author;  // Author ka naam store karega
    float price;    // Book ki price store karega
    int pages;      // Pages ki tadaad store karega
};

int main() {
    // Declare and initialize two structure variables
    // Yahan hum book1 aur book2 bana rahe hain aur values assign kar rahe hain
    Book book1 = {"C++ Basics", "Ali Khan", 500.50, 200};
    Book book2 = {"Programming Guide", "Ahmed Raza", 750.75, 350};

    cout << "Book 1 Details:" << endl;
    cout << "Title: " << book1.title << endl;    // Title print karega
    cout << "Author: " << book1.author << endl;    // Author print karega
    cout << "Price: " << book1.price << endl;    // Price print karega
    cout << "Pages: " << book1.pages << endl;    // Pages print karega

    cout << endl;

    // Display values of book2
    // book2 ki values print kar rahe hain
    cout << "Book 2 Details:" << endl;
    cout << "Title: " << book2.title << endl;    // Title print karega
    cout << "Author: " << book2.author << endl;    // Author print karega
    cout << "Price: " << book2.price << endl;    // Price print karega
    cout << "Pages: " << book2.pages << endl;    // Pages print karega

    return 0; }
```